The Review Process

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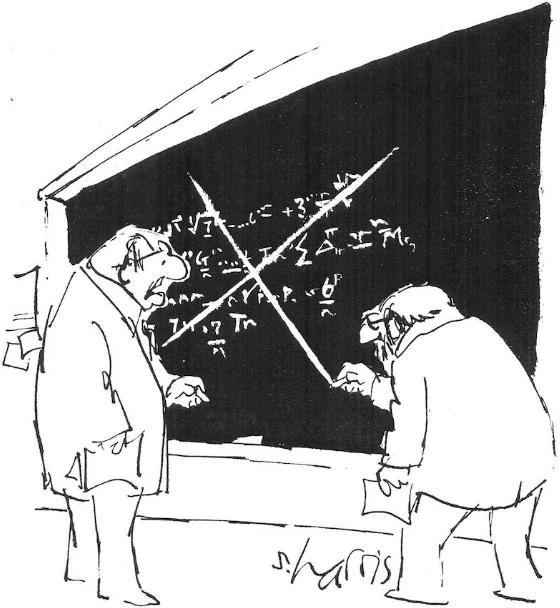


"THAT'S IT? THAT'S PEER REVIEW?"

Outline

The review process

Structuring reviews

Poster design

From last time ...

Publication procedure:

1. Scientific work according to proper methods
2. Document your methods and results in paper
3. Submit paper to a relevant conference (or journal)
4. Receive reviews and revise your papers accordingly
5. Re-submit revised paper
6. Publication

Also from last time ...

1. Do your scientific work
2. Write a paper on it by November 27
3. Read your fellow students' papers and write reviews by December 6
4. Receive the reviews and revise your papers accordingly
5. SEMCON December 19
6. Exam in January - poster, paper AND reviews!

Timeline

- November 27: Submit draft papers to your fellow groups
- December 6: Return reviews to authors
- December 16: Submit poster file to printer
- December 17: Submit title and 400-word abstract of revised paper
- December 17: Upload project to Digital Exam (article+worksheets)
- December 19: SEMCON
- Exam in January - poster, paper AND reviews!

Why Should You be a Reviewer?

Moral obligation (Kant's categoric imperative):

- When you wish your papers to be reviewed, you should also be willing to do reviews yourself

Give yourself insight in 'hot' subjects:

- You see papers possibly several years before their publication (although you can not use this directly in your own research)

Broaden your mind:

- Most likely, you will be reading papers that are slightly outside your main field. This might give 'bridging' ideas

Are You an Appropriate Reviewer?

When you receive a manuscript, ask yourself immediately:

- Are you reasonably familiar with the subject?
- Do you know the references that provide the background?

If you know some parts of the subject, you might restrict attention to these parts, but this should be mentioned in the review (or in comments to the editor)

If you are not familiar with the subject at all, you should return the manuscript immediately, possibly with suggestions for better qualified reviewers.

What Are the Objectives of Your Review?

The objectives of a peer reviewer is mainly to determine:

- novelty
- correctness
- significance

of the manuscript submitted.

How is Novelty Determined

A manuscript is not publishable if its results are already published, but

- this only concerns *research* papers; survey or overview papers has the 'opposite' objective
- usually, the author and/or reviewer can be held responsible only for results published in peer reviewed journals and conference proceedings, and not for e.g. doctoral dissertations published several years prior to present manuscript
- it is fair to judge the paper relative to papers published at submission time

How is Correctness Determined

Standards of correctness vary highly within different fields of science; hence the assessment on correctness should be carried out relative to the scientific tradition in the relevant field/journal/conference.
For 'theorem/proof' papers:

- check the proof!

For papers with simulation studies:

- check if data are provided that facilitate reproduction of results

It is usually not expected from the reviewer to redo the simulations.

For papers based on experimental studies:

- check if data are provided that allow experiments to be reproduced

It might be necessary to assess correctness of references!

How is Significance Determined

On one hand consider

- Does the paper contain original concepts?
- Does the paper contain creative concepts?
- Does the paper contain unusual concepts?

On the other hand consider

- Are the results just minor extensions of existing results?
- Do you anticipate that the results could lead to further research and/or applications?

As a rule of thumb, it is dangerous to reject a paper for lack of significance only, if there are no hard, objective arguments for this.

Be Specific and Helpful

Substantiate your claims in a useful way! Whenever possible, you should provide examples for your claims, to demonstrate to the editor that you are impartial and objective.

If you claim lack of clarity:

- give example of unclear paragraphs in the manuscript

If you claim lack of rigor:

- give examples for technical improvements

If you claim lack of novelty:

- provide references that overlap with the manuscript

If you claim lack of correctness:

- find a counter example!

Respect Confidentiality

As a reviewer you are anonymous. The peer reviewing system relies on the assumptions that you:

- do not use the results in your own research (before publication)
- do not disclose the results to other colleagues
- resist any temptation to push a *unjustified* reference to your own work on the author(s)

If you wish to use or cite the results before publication, you have to reveal your identity to the authors and ask their permission after the review process has been completed. The authors are not obliged to accept your plea.

Be Objective and Fair

You are anonymous, but you should

- formulate criticism as you would to the authors' face
- only exert criticism to a standard you fulfill in your own work (Kant)
- respect different points of view on how to approach a class of problems!

In practice, a review will often convey all the weak points of a paper. However, the review will be received by an author who struggled to present her/his research result in the hope that it would be well received and appreciated. To that end, you could also consider to include positive comments!

Some questions to ask oneself when doing a rev

- What is the purpose of the paper?
- Is the paper appropriate for the venue (conference/journal)
- Is the goal of the paper significant?
- Is the method of approach valid?
- Is the execution of the method/research correct?
- Are the conclusions drawn from the results valid?
- Is the overall presentation satisfactory?
- What did you, the reader, learn?

Outline

Structure of reviews

Overall assessment: In this part (typically one or two paragraphs) you can give an overview of what you considered strong and weak points, and what you considered to be the main message(s).

General comments: Here you can give comments on the structural level, either with respect to contents or paper structure. You can use a numbering scheme, e.g.:

GC1: ...
GC2: ...

Specific comments: This part typically would contain rather detailed comments, such as minor mistakes, missing explanation of terminology, etc., often referring to specific lines in the paper. Again, it could be a good idea to use a numbering scheme:

SC1: ...
SC2: ...

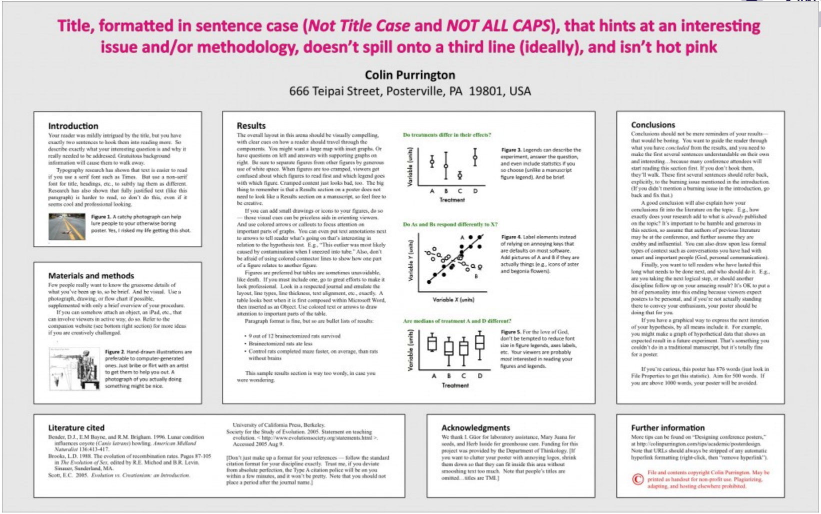
Revision Report

If a revision report is required or in place:

- If the editor gives specific instructions, one section should be dedicated to a response to these.
- For each suggestion of the reviewers, a response must be given. A good structure is to cite the suggestion followed by your response.
- As a general rule, try to follow the advice of the reviewers as much as possible and probably a bit more ...
- Try to argue with the reviewers only if you honestly believe that your arguments will convince the editor and the reviewers.

Outline

Poster



http://colinpurrington.com/tips/academic/posterdesign

Elements on a poster

- (Short) title and authors
- An introduction to your burning question
- An overview of your novel approach
- Your amazing results in graphical form
- Some insightful discussion of aforementioned results
- A *brief* listing of important literature
- A *brief* acknowledgement of the tremendous assistance and financial support conned from others

Readability!

- Not too much text
- Follow the IMRAD structure
- Consistent color scheme
- Use figures as much as possible
- Print the poster in A4; you should be able to read it at arm's length

Bad example

http://colinpurrington.com/tips/academic/posterdesign

http://colinpurrington.com/tips/academic/posterdesign

Horrible posters are?

- unorganized
- cluttered
- extremely confusing to those outside your field of study
- filled with superfluous text
- picture-less
- embarrassing to your colleagues, your university, and yourselves.

http://http://justinlmatthews.com/posterhelp/posterguide/

Presenting the poster

- Do not chew tobacco; do not chew gum
- Keep your hands out of your pockets
- Do not wear sunglasses indoors
- Do not refer to notes when explaining your poster
- Speak to your viewers as you explain your poster. I.e., do not talk to your poster
- Avoid vagueness such as “this figure shows our main result.” Explain your points concisely and in the necessary detail

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A final word on SEMCON

- Two presentation sessions run in parallel in the morning
- You will have **15 minutes** for the presentation followed by 5 minutes for questions
- Send your presentations in PDF or PowerPoint by email to Rasmus the day before SEMCON - state the name of the presenter
- Similarly, there will be two parallel poster sessions in the afternoon
- Poster speeches must be given in **three minutes**